

Approach

The answer is **always computed by a deterministic symbolic solver**, never by a language model. The LLM (open-source, **Qwen2.5-3B-Instruct**, Apache-2.0, $\leq 8B$ params) is confined to three utility roles: (1) extract physical quantities from prose when regex misses, (2) translate a natural-language claim into first-order logic, (3) phrase the final explanation. Every numeric result comes from SymPy/pint; every logical verdict from forward-chaining or a Z3 proof. This makes the system *verifiable*: the explanation is generated from the solver’s own trace, so it cannot diverge from the answer – directly serving the rubric’s P2 (explanation) and P3 (reasoning depth).

Pipeline

- **Router**: presence of **premises**-NL $\rightarrow$ logic, else physics.
- **Physics**: question-cleaner $\rightarrow$ topic classifier (ID-prefix + keyword + symbol fallback) $\rightarrow$ quantity extractor (regex; LLM fallback) $\rightarrow$ unit normalisation (pint) $\rightarrow$ formula registry (24 formulas, YAML $\rightarrow$ SymPy, incl. Coulomb vector superposition) $\rightarrow$ solve $\rightarrow$ unit/sanity verifier $\rightarrow$ trace explanation.
- **Logic**: Jaccard premise selector $\rightarrow$ rule parser $\rightarrow$ forward chainer $\rightarrow$ **Z3** entailment (multi-variable universals, comparison literals) when surface reasoning is inconclusive $\rightarrow$ per-option entailment for MC; emits **premises**: ["P1", ...], the FOL, and a step CoT.
- **Self-correction** (deployed, ON): the solver re-checks the LLM draft; feedback is *solver-grounded* (never invents numbers/premises), ≤ 2 rounds.
- Output schema: {**answer**, **explanation**, **cot**, **premises**, **fol**, **confidence**}; malformed JSON is impossible (Pydantic) and the LLM path degrades to the deterministic answer if the model is unreachable (never a 500).

Training (ablation, not on the critical path)

We QLoRA-SFT’d Qwen2.5-7B purely to stabilise output *format* (ADR 0006/0013). Measured impact was marginal (physics +1.5 pp, logic flat) at $\sim 5\times$ latency: **the system is solver-bound, not LLM-bound**. The deployed endpoint therefore omits the adapter and runs the small 3B extractor for latency; the adapter is retained as a documented, reproducible ablation.

Data & compliance

- **Only cleaned_exact_dataset_package** is used (no external data/models).
- LLM is open-source $\leq 8B$; **no GPT/Claude/Gemini** anywhere in the pipeline.
- Eval = strict 10% holdout (seed 42); a frozen unit-/round-aware scorer (ADR 0009/0011/0012). 228 automated tests; ruff+mypy CI.

Internal results (10% holdout, frozen scorer)

Task	Rule-only	Deployed system	Lift
Physics (exact value+unit)	0.8%	28.6%	$\sim 36\times$
Logic (exact label)	35.9%	37.5%	+1.6 pp

Physics correctness is reported under a deliberately strict exact value-and-unit match; the dominant residual is LLM extraction recall on hard multi-step problems, not solver error (the solver is exact where extraction succeeds). Logic is translator-bound: the symbolic core (Z3) is sound; gains track NL $\rightarrow$ FOL quality.

Deployment

`docker compose up` $\rightarrow$ GPU Ollama sidecar + CPU FastAPI solver; always-on with `/healthz`, model pinned resident, cron pre-warm, snapshot backup. Reproduce: `uv run pytest -q; uv run python scripts/run_eval.py --task {physics,logic} --with-llm`.